

MySQL School Examination System - 50 Practice Problems

Dataset Setup + Easy, Medium and Hard Problems (Solutions Not Included)

Database Setup

```
CREATE TABLE Students ( id INT PRIMARY KEY, roll_number INT UNIQUE NOT NULL,
student_name VARCHAR(100) NOT NULL, class VARCHAR(10) NOT NULL ); CREATE TABLE
Subjects ( id INT PRIMARY KEY, subject_name VARCHAR(100) NOT NULL ); CREATE TABLE
Scores ( std_id INT, sub_id INT, marks INT, PRIMARY KEY(std_id, sub_id), FOREIGN
KEY(std_id) REFERENCES Students(id), FOREIGN KEY(sub_id) REFERENCES Subjects(id) );
INSERT INTO Students VALUES (1,101,'Rahul Sharma','9A'), (2,102,'Priya Verma','9A'),
(3,103,'Amit Patel','9B'), (4,104,'Neha Gupta','10A'), (5,105,'Karan Singh','10B'),
(6,106,'Riya Shah','10B'); INSERT INTO Subjects VALUES (1,'Mathematics'),
(2,'Science'), (3,'English'), (4,'Social Studies'), (5,'Computer'); INSERT INTO
Scores VALUES (1,1,85),(1,2,78),(1,3,92),(1,4,81),(1,5,88),
(2,1,65),(2,2,72),(2,3,75),(2,4,68),(2,5,70),
(3,1,95),(3,2,91),(3,3,89),(3,4,93),(3,5,97),
(4,1,58),(4,2,61),(4,3,66),(4,4,55),(4,5,60),
(5,1,82),(5,2,84),(5,3,79),(5,4,88),(5,5,90);
```

Easy Problems (15)

1. Display all students.
2. Display all subjects.
3. Display all score records.
4. Display students from class 9A.
5. Display students from class 10B.
6. Display student having ID 3.
7. Display subject having ID 2.
8. Display score records having marks greater than 80.
9. Display score records having marks less than 60.
10. Display score records having marks between 70 and 90.
11. Display students sorted by name.
12. Display subjects sorted alphabetically.
13. Display score records sorted by marks ascending.
14. Display score records sorted by marks descending.
15. Display all students whose name starts with 'R'.

Medium Problems (20)

16. Find students who scored above 90 in any subject.
17. Find students who scored below 60 in any subject.
18. Find students who appeared in Computer subject.
19. Find subjects having at least 4 score entries.
20. Find students having marks between 75 and 85 in any subject.
21. Find the highest mark recorded in the database.
22. Find the lowest mark recorded in the database.
23. Find average marks for each subject.
24. Find total marks scored by each student.
25. Find students whose total marks exceed 400.
26. Find students who appeared in all subjects.
27. Find subjects where average marks exceed 80.
28. Count students in each class.
29. Count score records for each subject.
30. Find students having more than 3 score entries.
31. Find subjects having maximum marks above 90.
32. Find students whose lowest mark is below 70.
33. Find students whose highest mark is above 90.
34. Find classes containing more than one student.
35. Find students who never scored below 70.

Hard Problems (15)

36. Find students whose total marks are above the average total marks of all students.
37. Find subjects whose average marks are below the overall average marks.
38. Find the subject having the highest average marks.
39. Find the subject having the lowest average marks.
40. Find students who scored higher than the maximum score in Mathematics.
41. Find students who have exactly one score record.
42. Find subjects that never received a score above 80.
43. Find students whose highest and lowest marks differ by at least 15.
44. Find the second-highest mark in the database.
45. Find all records having the second-highest mark.
46. Find students who scored below 75 in more than one subject.
47. Find subjects having the same number of score entries as Science.
48. Find students who have never appeared for any examination.
49. Find top 3 students based on total marks.
50. Find subjects having at least three score entries and display them in descending subject-name order.